Cambridge Mathematics Practice

Stage 8 / Grade 7 – Applied Percentages Worksheets

Name: _____ **Date:** _____
Class: _____ **Score:** _____

Instructions

- Answer all questions clearly in the spaces provided or on separate paper.
- Show all steps in your working to obtain full marks.
- Give non-exact numerical answers correct to 3 significant figures or 1 decimal place for percentages, unless stated otherwise.
- Calculators are permitted unless explicitly instructed.

Questions

1. A sports store increased the price of a tennis racket from \$80 to \$96. Calculate the percentage increase in the price of the racket.

2. During a end-of-year sale, a smart watch originally priced at \$240 is offered with a 15% discount.

(a) Calculate the total discount amount in dollars.

(b) Find the final sale price of the smart watch.

3. A pair of running shoes is sold for \$68 after a 20% discount was applied to the original tagged price. Calculate the original price of the running shoes before the discount.

4. Oliver deposits \$1,800 into a savings account that pays simple interest at a rate of 4% per annum. Calculate the total amount of money in his account after 3 years.

5. A bookshop owner buys a set of novels for £40 and sells it later for £52. Calculate the percentage profit made by the shop owner.

6. The final price of an electronic gadget including a 20% Value Added Tax (VAT) is £156. Calculate the cost of the gadget before VAT was added.

7. The total student enrolment at Greenfield Academy was 1,200 in the year 2022.

(a) Between 2022 and 2023, enrolment increased by 5%. Calculate the number of students in 2023.

(b) Between 2023 and 2024, enrolment decreased by 8%. Calculate the number of students in 2024.

8. In a survey involving 400 school students:

(a) 35% stated that mathematics is their favorite subject. Calculate how many students chose mathematics.

(b) $\frac{5}{8}$ of the remaining students chose science. Calculate the percentage of the total survey group who chose science.

9. Emma receives a monthly salary of \$4,200. She spends 35% on rent, 20% on food and utilities, and saves $\frac{1}{3}$ of the remaining money.

(a) Calculate the total amount Emma spends on rent, food, and utilities combined.

(b) How much money does Emma save each month?

10. Liam invests \$3,000 in a fixed deposit account offering compound interest at a rate of 5% per annum. Calculate the total amount of money in his account after 2 years.

11. Mia claims: "If an item increases in price by 10% and then decreases by 10%, the final price returns to the original price."

(a) Show by mathematical calculation, starting with an initial price of \$100, that Mia's claim is incorrect.

(b) Calculate the overall percentage change in price relative to the original amount.

12. After an upgrade in production efficiency, a factory's daily output increased by 25% to 6,500 items per day. Calculate the daily output before the efficiency upgrade.

13. A new motorcycle was purchased for £8,500. It depreciates at a rate of 10% per year.

(a) Calculate the value of the motorcycle after 1 year.

(b) Calculate the value of the motorcycle after 3 years. Round your answer to the nearest pound.

14. A collector bought a rare comic book for \$450 and sold it three years later for \$378. Calculate the percentage loss incurred.

15. A rectangular school yard measures 90 metres by 60 metres. The maintenance team covers 75% of the total area with fresh grass turf. Calculate the area of the yard, in square metres, that is **not** covered with grass turf.

16. Shop A and Shop B sell the same model of smartphone, which has a list price of \$600.

- **Shop A** offers a straight discount of 30%.
- **Shop B** offers an initial price reduction of \$100 followed by an additional 20% off the reduced price.

Determine which shop offers the lower final price and state the difference in price.

17. An architect estimated the height of a building to be 45 metres. Upon construction, the actual height was measured to be 48 metres. Calculate the percentage error of the architect's estimate.

18. A metal alloy weighing 600 grams consists of 55% copper, 30% zinc, and the remaining portion is nickel.

(a) Calculate the mass of nickel present in the alloy, in grams.

(b) Express the ratio of copper to zinc in its simplest integer form.

19. Lucas earns a gross annual salary of £36,000.

(a) The first £12,500 of his income is tax-free. Calculate his taxable income.

(b) He pays income tax at a rate of 20% on all taxable income. Calculate the total annual income tax he pays.

(c) Express his tax payment as a percentage of his total gross income.

20. A bakery produced 8,000 muffins in October.

(a) In November, production increased by 15%. How many muffins were produced in November?

(b) In November, 2.5% of the muffins produced were rejected due to quality standards. Calculate the number of accepted (non-defective) muffins in November.

(c) If each accepted muffin is sold for $2.50, calculate the total revenue generated from November's accepted muffins.

Answer Key & Solutions

Cambridge Grade 7 – Applied Percentages

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1. **Price Increase:** $\$96 - \$80 = \$16$
   Percentage Increase $= \frac{16}{80} \times 100\% =$ **20%**

2. (a) Discount $= 15\% \times 240 = 0.15 \times 240 = \$$**36**
   (b) Sale Price $= 240 - 36 = \$$**204**

3. Percentage of Original Price $= 100\% - 20\% = 80\%$
   Original Price $= \frac{68}{0.80} = \$$**85**

4. Simple Interest $= I = P \times r \times t = 1800 \times 0.04 \times 3 = \$216$
   Total Amount $= 1800 + 216 = \$$**2, 016**

5. Profit $= \$52 - \$40 = \$12$
   Percentage Profit $= \frac{12}{40} \times 100\% =$ **30%**

6. Cost including VAT $= 120\% = 1.20$
   Cost before VAT $= \frac{156}{1.20} =$ ***$*130**

7. (a) Enrolment 2023 $= 1200 \times 1.05 =$ **1, 260** students
   (b) Enrolment 2024 $= 1260 \times (1 - 0.08) = 1260 \times 0.92 =$ **1, 159.2** $\approx$ **1, 159** students

8. (a) Students choosing Math $= 35\% \times 400 = 0.35 \times 400 =$ **140** students
   (b) Remaining students $= 400 - 140 = 260$
   Students choosing Science $= \frac{5}{8} \times 260 = 162.5$
   Percentage of Total $= \frac{162.5}{400} \times 100\% =$ **40.625%** $\approx$ **40.6%**

9. (a) Total % spent $= 35\% + 20\% = 55\%$
   Total spent $= 55\% \times 4200 = 0.55 \times 4200 = \$$**2, 310**
   (b) Remaining salary $= 4200 - 2310 = \$1,890$
   Savings $= \frac{1}{3} \times 1890 = \$$**630**

10. Compound Interest Amount $= A = P(1+r)^n = 3000 \times (1+0.05)^2 = 3000 \times 1.1025 = \$$**3, 307.50**

11. (a) Price after 10% increase $= 100 \times 1.10 = \$110$
    Price after 10% decrease $= 110 \times 0.90 = \$99$
    Since $\$99 \neq \$100$, Mia's claim is **incorrect**.
    (b) Overall change $= 99 - 100 = -\$1$
    Percentage change $= \frac{-1}{100} \times 100\% = -$**1%** (1% decrease)

12. Output before upgrade $= \frac{6500}{1+0.25} = \frac{6500}{1.25} =$ **5, 200** items

13. (a) Value after 1 year $= 8500 \times (1 - 0.10) = 8500 \times 0.90 =$ ***$*7, 650**
    (b) Value after 3 years $= 8500 \times (0.90)^3 = 8500 \times 0.729 = \$6,196.50 \approx$ ***$*6, 197**

14. Loss $= \$450 - \$378 = \$72$
    Percentage Loss $= \frac{72}{450} \times 100\% =$ **16%**

15. Total Area $= 90 \times 60 = 5,400$ m^2
    Percentage not covered $= 100\% - 75\% = 25\%$
    Area not covered $= 25\% \times 5400 = 0.25 \times 5400 =$ **1, 350** m^2

16. Shop A Final Price $= 600 \times (1 - 0.30) = 600 \times 0.70 = \$\mathbf{420}$
Shop B Initial Reduction $= 600 - 100 = \$500$
Shop B Final Price $= 500 \times (1 - 0.20) = 500 \times 0.80 = \$\mathbf{400}$
Conclusion: Shop B offers the lower price by $\$\mathbf{20}$ ($\$420 - \$400$).

17. Percentage Error $= \frac{|\text{Estimated} - \text{Actual}|}{\text{Actual}} \times 100\% = \frac{|45-48|}{48} \times 100\% = \frac{3}{48} \times 100\% = \mathbf{6.25}\%$

18. (a) Percentage of Nickel $= 100\% - (55\% + 30\%) = 15\%$
Mass of Nickel $= 15\% \times 600 = 0.15 \times 600 =$ **90** grams
(b) Ratio Copper : Zinc $= 55 : 30 = \frac{55}{5} : \frac{30}{5} =$ **11** : **6**

19. (a) Taxable Income $= 36000 - 12500 = \mathbf{\$23,500}$
(b) Total Tax $= 20\% \times 23500 = 0.20 \times 23500 = \mathbf{\$4,700}$
(c) Percentage of Gross Income $= \frac{4700}{36000} \times 100\% \approx \mathbf{13.06}\% \approx \mathbf{13.1}\%$

20. (a) November Production $= 8000 \times (1 + 0.15) = 8000 \times 1.15 =$ **9**, **200** muffins
(b) Accepted Percentage $= 100\% - 2.5\% = 97.5\%$
Accepted Muffins $= 9200 \times 0.975 =$ **8**, **970** muffins
(c) Total Revenue $= 8970 \times \$2.50 = \$\mathbf{22,425}$